

5.1.1 Machine Learning (ML)

مشین لرننگ ماڈل وہ نظام ہے جو ڈیٹا سے سیکھ کر نتیجہ دیتا ہے — کمپیوٹر پروگرام کا تربیت یافتہ دماغ۔

CS XII

Chapter 5: Application and Impacts of Computing

Lecture 02 of 15

5.1.1 Machine Learning (ML)

LEARN IT

A **machine learning model** is the system that **learns from data and gives an output**. Think of it as the **trained brain** of a computer program.

In ordinary programming you write every rule yourself: *if marks ≥ 40 then "Pass"*. In machine learning you **do not list every rule**. You give the model many examples (data), it finds the patterns, and later it decides about new cases it has never seen.

How it works — three steps:

1. Data input → 2. Learning (find patterns) → 3. Prediction on new data

1. **Data input.** The model is given massive amounts of data — for example thousands of past BISE marksheets, or millions of crop photos from Sindh fields.
2. **Learning.** It studies the data and **learns patterns and rules automatically**. It might notice that students who scored low in Chapter 3 also failed the board exam, or that yellow spots on cotton leaves often mean a pest.
3. **Prediction.** After training, it can make **accurate predictions or decisions** when given **new, unseen data** — a new student's record, or a new leaf photograph.

Note from the source: there are many types of machine-learning models. A **neural network** is one important type. **Deep learning** is a highly advanced method of machine learning.

WORKED EXAMPLE

Worked example: a wheat-yield model for a farm near Sukkur.

Step 1 — Data input. For 8 years the farm recorded: rainfall (mm), temperature, fertiliser bags used, and the actual yield (maunds per acre).

Step 2 — Learning. The model studies those rows. It finds, for example, that yield rises with fertiliser up to a point, and falls when temperature is very high in April.

Step 3 — Prediction. This year the farmer types in this season's rainfall and temperature. The model has never seen this exact year before, but it predicts "about 32 maunds per acre." The farmer plans storage and a buyer in advance.

Nobody wrote a giant list of "if rain = 120 and temp = 38 then yield = ...". The model learned the pattern from data.

REMEMBER IT

D-L-P. Machine learning is **Data** in, **Learn** patterns, **Predict**. "Data Learns to Predict."

COMMON MISTAKE

Mistake: "The computer is intelligent on day one."

Fix: An ML model is useful only **after training** on enough good data. Garbage data → garbage predictions. Also, neural networks are *one* type of ML, not the only type; deep learning is a *method* that uses deep neural networks.

اردو میں سمجھیں

مشین لرننگ ماڈل وہ نظام ہے جو ڈیٹا سے سیکھ کر نتیجہ دیتا ہے — کمپیوٹر پروگرام کا تربیت یافتہ دماغ۔ پہلے بہت سا ڈیٹا دیا جاتا ہے، پھر ماڈل خود اصول اور پیٹرن سیکھتا ہے، اور تربیت کے بعد نئے ان دیکھے ڈیٹا پر پیش گوئی کرتا ہے۔ نیورل نیٹ ورک مشین لرننگ کی ایک اہم قسم ہے، اور ڈیپ لرننگ اس کا ایک اعلیٰ طریقہ ہے۔

CHECK YOURSELF

1. What are the three steps of how an ML model works?
2. What is a machine-learning model compared to in the source?
3. After training, what kind of data can the model handle?

Answers: 1. Data input, learning, prediction. 2. The trained brain of a computer program. 3. New, unseen data.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Machine learning lets a model:

- a. Improve from examples, not only hard-coded rules
- b. Run without data
- c. Replace all maths
- d. Ignore errors always

MCQ 2. Training data is:

- a. Examples used to fit the model
- b. The final user password
- c. OSI headers
- d. Figma frames

MCQ 3. A model that memorizes noise is:

- a. Overfitting
- b. Perfect science
- c. A stack
- d. A pie

MCQ 4. Supervised learning uses:

- a. Labels (the right answers)
- b. No data
- c. Only unsupervised always
- d. Only cables

MCQ 5. ML in this book is the “trained brain” metaphor for:

- a. Mapping inputs to outputs after seeing examples
- b. A DBMS trigger
- c. A foreign key
- d. A sprint

MCQ 6. Garbage data yields:

- a. Garbage predictions
- b. Perfect fairness
- c. $O(1)$ time always
- d. Free RAM

Extra short questions

1. Define ML in the book’s words/your own accurate line.
2. Give two applications.
3. What is a label?
4. One ethical issue.

Extra long question

1. Explain ML to a principal who wants to predict drop-outs: data, train, test, a risk, a human check.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. A model that learns from data to produce outputs without being explicitly programmed for every case.
2. Spam; recommendations; marks prediction; vision.
3. The known answer attached to a training example.
4. Bias in the training data is copied into decisions.

Long

1. Past attendance+marks → model → test on held-out year. Risk: punishing poverty correlated with absence. Human counsellor decides, model only flags.

Next lecture: 5.1.2 Neural Networks (NN)